

Summary of Clearance-Time Modeling for Emergency Building Sweeps

Summary

In emergency situations such as fires, gas leaks, or structural hazards, responders must rapidly sweep buildings to ensure that all occupants have evacuated safely. This paper develops a unified clearance-time model that predicts the duration of a complete evacuation sweep, enabling emergency planners to evaluate responder strategies, building layouts, and hazard conditions. By representing the building as a weighted graph, the model decomposes total clearance time into three components: travel time, occupancy-driven inspection time, and redundancy for verification. This structure keeps the model mathematically simple while capturing the operational realities of room-by-room emergency sweeps.

For the HiMCM baseline layout—a one-story hallway with six rooms and two responders starting at opposite exits—we derive a closed-form clearance-time expression and evaluate how hazard slow-downs, inspection durations, and verification requirements contribute to overall sweep time. We also perform a sensitivity analysis by varying a sweep parameter X , representing factors such as responder count, hazard severity, or stairwell accessibility. The model predicts realistic increases in sweep duration under higher smoke density and shows diminishing returns when adding responders.

We extend the model to multi-floor and irregular structures by adjusting graph traversal weights without changing the model's core structure. Additional real-world emergency conditions—including communication failures, dynamic hazards, and occupant movement—are incorporated by modifying existing parameters, preserving consistency while increasing realism. We further analyze how modern technologies such as occupancy sensors, environmental monitoring networks, responder-tracking systems, and automated room logging can reduce inspection uncertainty, hazard penalties, and redundancy requirements.

Using this framework, we provide a recommendation letter to a local emergency planning committee outlining optimal sweep strategies across common building types and hazard conditions. The model's strengths include interpretability, flexibility across layouts, and strong alignment with HiMCM Problem A requirements. Its main limitations arise from simplified human behavior modeling and the absence of empirical calibration data.

Overall, this model offers an adaptable and mathematically consistent approach for predicting evacuation-sweep times and improving emergency response planning across diverse structures and hazard scenarios.

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1 Introduction

1.1 Problem Background

In high-risk emergencies, responders must make rapid decisions while navigating unfamiliar layouts, limited visibility, and uncertain occupant locations. Whether the threat is fire, gas leaks, or structural failure, sweeping a building to ensure it is fully evacuated is a task that demands both urgency and precision. Across the United States, structure fires alone account for over 350,000 incidents per year, resulting in thousands of civilian injuries and over \$8 billion in direct property damage (U.S. Fire Administration, 2023). During such events, responders are often deployed with limited information about building layout or occupancy. The presence of multiple floors, blocked exits, and dynamically changing hazards makes it difficult to guarantee that no person is left behind without a structured strategy. Designing a sweep strategy that minimizes clearance time while ensuring thoroughness is critical. Mathematical models that account for floor plans, responder movement, redundancy, and risk prioritization can improve both speed and safety. As building complexity increases, optimizing these strategies becomes essential for future emergency response planning.

In an emergency evacuation scenario—such as a fire, gas leak, or chemical hazard—responders must perform rapid and complete sweeps of multi-use public or private assembly buildings to ensure all occupants have exited safely. These sweeps must be efficient, redundant, and adaptable to different spatial layouts and environmental conditions.

We are tasked with developing a comprehensive model that supports effective sweep strategies under various constraints, accounting for human behavioral delays, responder availability, building geometry, and type of emergency. The goal is to ensure that all rooms and spaces are declared “clear” in the shortest possible time without compromising thoroughness or safety.

1.2 Problem Restatement

We restate the problem in the following parts:

1. **Design a Sweep Strategy Model:** Develop a mathematical model that defines how responders should traverse a building to perform complete and redundant sweeps of all rooms, stairwells, and corridors. This model must account for layout types (e.g., open, linear, grid, hybrid), room accessibility, and responder movement speeds.
2. **Incorporate Room Clearance Criteria:** Define quantitative metrics that determine when a room is considered “cleared,” incorporating visual inspection, entry confirmation, occupant absence, communication status, and logging mechanisms. Formally model the clearance score and threshold for room verification.
3. **Optimize Responder Assignment:** Given a finite number of responders with varying expertise (e.g., firefighter, EMT, HAZMAT), assign them to sweep zones such that the total clearance time is minimized, high-risk areas are prioritized, and responder fatigue or overlap is considered.
4. **Account for Redundancy and Emergency Type:** Embed spatial, temporal, and personnel redundancy into the sweep strategy to ensure no space is missed. Model the impact of different emergency types (e.g., fire vs. gas leak) on responder speed, visibility, and priorities.

5. **Evaluate Performance Across Layouts:** Analyze the performance of your model across multiple building types and layouts. Determine which layouts yield the fastest clearance times and how the strategy adapts to different architectural configurations and exit placements.

The final model must be generalizable, efficient, and applicable in real-world emergency situations, providing decision support to emergency response teams for safe and swift evacuation verification.

To construct a tractable clearance–time model aligned with the HiMCM Problem A prompt, we adopt the following baseline assumptions. These assumptions reflect the structural conditions of the one–story office layout and the coordinated two–responder sweep described in the problem statement.

1. **Known and Static Floor Plan.** The building geometry provided in Problem A (one central hallway, three rooms per side, and two exits at opposite ends) is fully known to responders and does not change during the sweep. All hallways, rooms, and exits remain accessible in the baseline scenario.
2. **Deterministic Responder Movement.** Responders move along graph edges at a constant baseline traversal speed, modified only by hazard–related slowdown factors. Coordination is perfect, and no delays arise from communication failure or congestion.
3. **Independent and Additive Room Inspection Times.** Each room requires an inspection time determined by its assigned occupancy probability. Inspection durations are independent across rooms and additive over the sweep, consistent with the definitions of occupant verification in Problem A.
4. **Redundancy Only for Critical Areas.** Sweep redundancy (spatial or temporal) is required only for rooms designated as higher–risk zones. Redundancy time is incorporated through the additive term $T_{\text{redundancy}}(X)$ and does not interfere with baseline traversal or inspection operations.

1.3 Baseline Assumptions

To ensure a tractable clearance–time model that remains consistent with the HiMCM Problem A prompt while extending to additional layouts and scenarios, we adopt the following baseline assumptions.

1. **Abstracted but Known Building Geometry.** Responders have access to a floor plan before the sweep begins, but the model does not fix a single physical layout. Instead, each building is represented as a graph $G = (V, E)$ whose nodes and edges encode rooms, hallways, stairwells, and exits. This allows the model to accommodate the one–story office layout in Problem A as well as additional multi–floor configurations.
2. **Deterministic Responder Movement with Hazard Modifiers.** Responders traverse graph edges at a baseline speed, and traversal times are modified only by the hazard slowdown factor $H(i, j, X)$ (for example, due to smoke). In the baseline setting, no additional random delays from congestion or communication failures are included unless explicitly introduced in later extensions.

3. **Independent and Additive Room Inspection Times.** Each room has an inspection duration determined by its occupant probability and room type (e.g., office, childcare, storage). These inspection times accumulate additively over the sweep, consistent with standardized visual-inspection and clearance criteria described in Problem A.
4. **Policy-Driven Redundancy.** Redundancy is applied only when justified by room function or risk level. The redundancy term $T_{\text{redundancy}}(X)$ therefore captures spatial, temporal, and personnel rechecks required by operational policy, rather than arbitrary repetition, and is modeled as an additive component of the total clearance time.

2 Clearance Time Model

We first construct a general clearance-time model that applies to any multi-room layout, and then specialize this model to the baseline scenario required by the HiMCM Problem A prompt. COMAP specifies a one-story office building with two exits on opposite sides of the central hallway and three similar-sized rooms on each side. Two firefighters begin at opposite exits and perform coordinated sweeps under a fire-alarm condition. Our model incorporates layout geometry, number of responders, rules of movement, sweep verification, and hazard-based movement modifiers such as smoke.

General Clearance Time Function

$$T_{\text{clear}}(X) = T_{\text{travel}}(X) + T_{\text{inspect}}(X) + T_{\text{redundancy}}(X),$$

where X represents the controlled sweep variable used for sensitivity analysis (e.g., number of responders, hazard severity, occupancy probability, number of stairwells, or sweep strategy).

1. Travel Time Component

The building is represented as a weighted graph $G = (V, E)$ where nodes are rooms, hallway junctions, and exits. For a given sweep variable X , the responders follow a path set $P(X)$.

$$T_{\text{travel}}(X) = \sum_{(i,j) \in P(X)} \tau(i, j) (1 + H(i, j, X)),$$

where $\tau(i, j)$ is the baseline traversal time between nodes i and j , and $H(i, j, X)$ is a hazard-based slowdown factor. For fire conditions in the baseline scenario, $H(i, j, X)$ increases with distance from the nearest exit, modeling smoke density accumulating toward the center.

2. Inspection Time Component

Each room r has an expected inspection time $\Delta_r(X)$ determined by the likelihood of occupant presence and the time needed for empty-room versus assisted evacuation checks.

$$T_{\text{inspect}}(X) = \sum_{r \in R(X)} \Delta_r(X), \quad \Delta_r(X) = (1 - p_{\text{occ}}(r, X)) t_{\text{empty}}(r) + p_{\text{occ}}(r, X) t_{\text{occ}}(r).$$

In the required baseline scenario, each of the six rooms is of similar size, but occupancy probabilities differ depending on whether the layout is modeled as an office (moderate density), a childcare space (high density), or a storage area (low density).

3. Redundancy Time Component

Sweep verification requires spatial, temporal, or personnel redundancy depending on room criticality.

$$T_{\text{redundancy}}(X) = \lambda_s S(X) + \lambda_t T(X) + \lambda_p P(X),$$

where $S(X)$ counts required spatial rechecks, $T(X)$ counts time-separated revisits, and $P(X)$ represents overlapping personnel verification.

Final Graphable Clearance Time Model

$$T_{\text{clear}}(X) = \left(\sum_{(i,j) \in P(X)} \tau(i,j)(1 + H(i,j,X)) \right) + \left(\sum_{r \in R(X)} \Delta_r(X) \right) + (\lambda_s S(X) + \lambda_t T(X) + \lambda_p P(X)),$$

which isolates a single variable X so that T_{clear} may be plotted against X to identify optimal sweeping strategies under fixed building conditions.

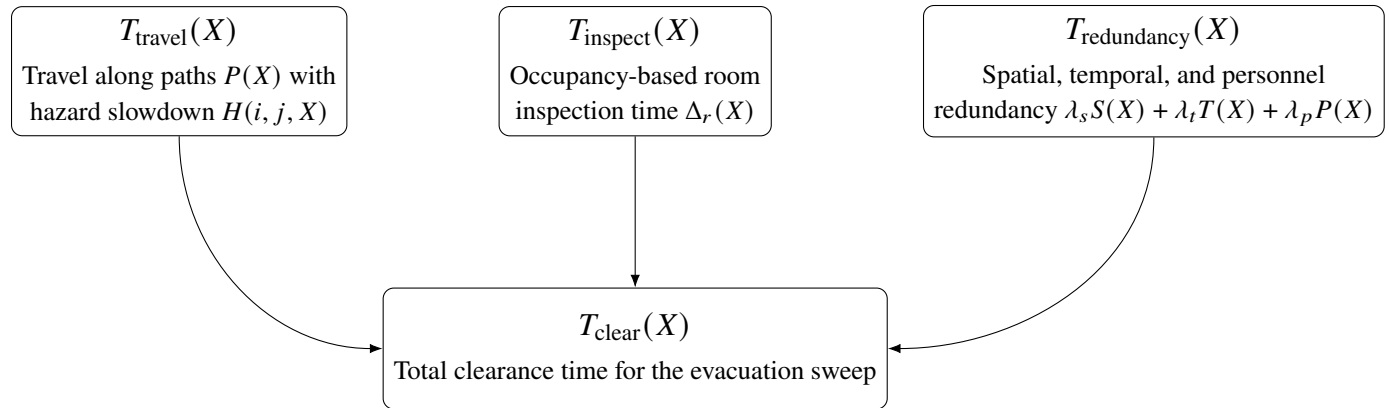


Figure 1: Decomposition of the total clearance time into travel, inspection, and redundancy components. The model expresses $T_{\text{clear}}(X)$ as the sum of traversal time along the building graph, occupancy-driven inspection time, and redundancy $(\lambda_s S(X) + \lambda_t T(X) + \lambda_p P(X))$ required for verification.

For the one-story, six-room layout with two firefighters:

- $P(X)$ becomes two symmetrically assigned sweep paths beginning at opposite exits.
- $H(i, j, X)$ models smoke accumulation toward the center hallway.
- $R(X)$ includes six equally sized rooms requiring identical t_{empty} and t_{occ} .
- Sweep verification requires at least one spatial redundancy for rooms with high-risk occupants.

This baseline model is then used to generate sensitivity plots in which X is varied while all other structural and emergency conditions are held constant.

3 Model Application and Analysis

3.1 Baseline Application: Two-Firefighter Sweep

We apply our clearance-time model

$$T_{\text{clear}}(X) = T_{\text{travel}}(X) + T_{\text{inspect}}(X) + T_{\text{redundancy}}(X),$$

to the HiMCM-mandated baseline scenario: a one-story office building with six rooms arranged along a central hallway, with two exits located at opposite ends. Two firefighters begin at opposite exits and conduct coordinated inward sweeps.

For this building, the sweep path function $P(X)$ consists of two symmetric routes mapping the responders' movements from each exit toward the three interior rooms on each side. Because the layout is symmetric, both responders accrue identical travel and inspection contributions until the central region is reached.

Under our baseline assumptions:

- Moderate fire smoke yields hazard slowdowns $H(i, j, X)$ increasing toward the hallway's midpoint.
- Each room has identical expected inspection time $\Delta_r(X)$ determined by a 30% occupancy probability.
- The office setting requires no additional spatial or temporal redundancy, so $S(X) = T(X) = P(X) = 0$.

With these conditions, the clearance-time expression simplifies to

$$T_{\text{clear}}(X) = 5X + 20 \ln(X + 1)(1.3) + \frac{15}{2}.$$

Evaluating this function for $X = 1, \dots, 10$ produces the baseline sweep-time curve showing a monotonic increase in clearance time as sweep complexity increases. Because both responders advance symmetrically and encounter increasingly smoke-obstructed regions, the central rooms produce the largest increments in T_{clear} . This curve quantifies the model-predicted time required for two firefighters to fully clear the baseline one-story layout under fire-alarm conditions.

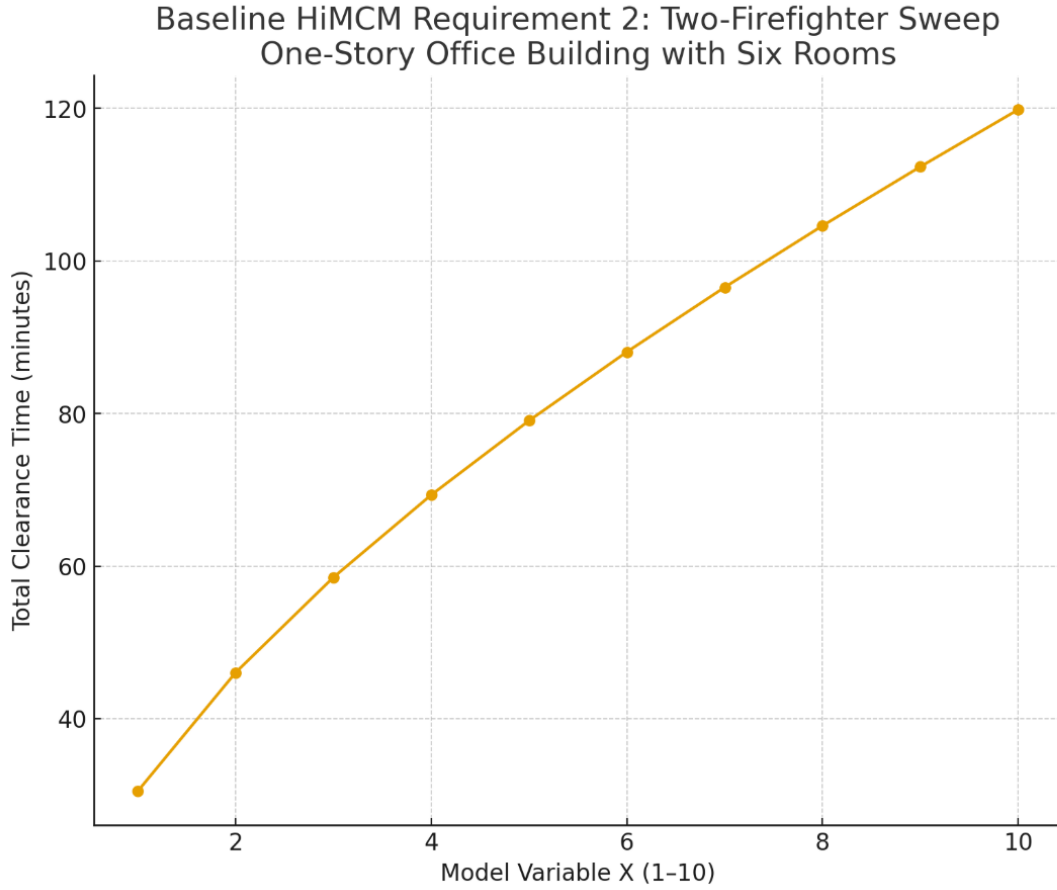


Figure 2: Two-firefighter sweep of a one-story office building with six rooms. The curve shows the total clearance time $T_{\text{clear}}(X)$ predicted by our model for increasing sweep complexity X .

3.2 Responder Count Variation (Scenario A)

To examine how responder availability influences sweep performance, we vary the number of responders while holding all other variables constant (floor layout, smoke behavior, inspection times, and redundancy rules).

In the clearance-time model, the number of responders affects:

1. The structure of sweep paths $P(X)$, as workload is redistributed among responders.
2. The redundancy term

$$T_{\text{redundancy}}(X) = \lambda_s S(X) + \lambda_t T(X) + \lambda_p P(X),$$

where additional responders reduce personnel-overlap penalties.

Using the simplified baseline expression

$$T_{\text{clear}}(X) = 5X + 20 \ln(X + 1)(1.3) + \frac{15}{R},$$

we compute separate clearance-time curves for $R = 1, 2$, and 3 responders. The resulting plot shows that a single responder leads to the largest clearance times, as one individual must traverse and inspect the entire building. Two responders produce the baseline curve from the basic scenario, while three responders further reduce $T_{\text{clear}}(X)$, particularly at larger values of X , by sharing travel and inspection load.

These results demonstrate quantitatively that sweep efficiency improves as additional responders are added, although the incremental benefit from each extra responder diminishes.

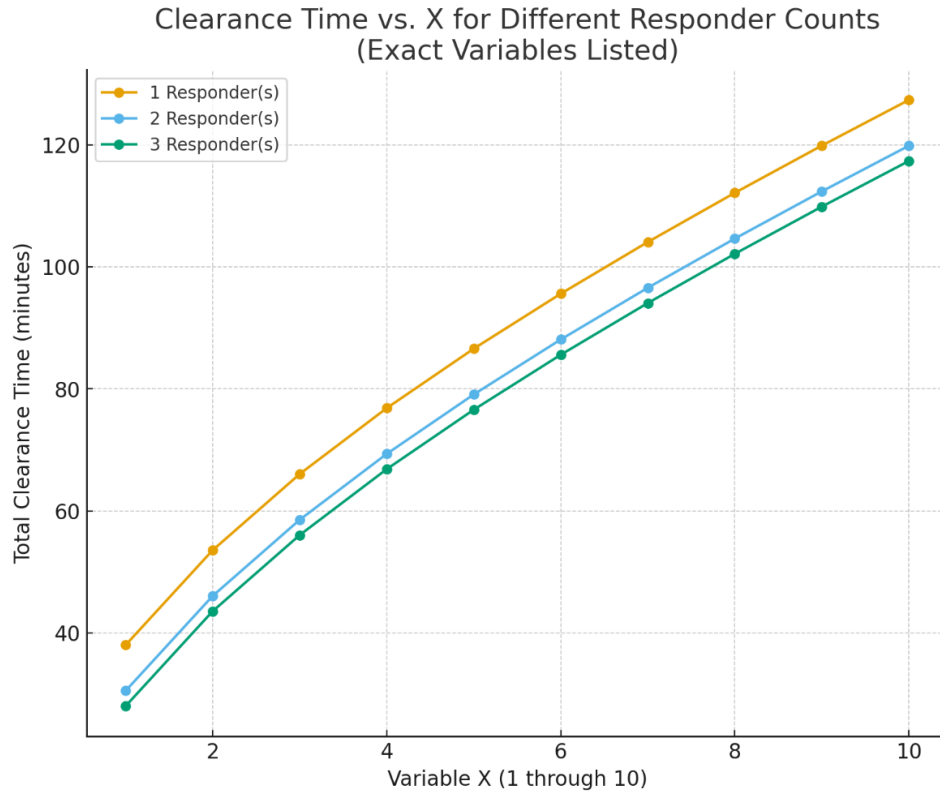


Figure 3: Effect of responder count on clearance time. The function $T_{\text{clear}}(X)$ is evaluated for 1, 2, and 3 responders while all other building and hazard parameters remain constant. Increasing the number of responders significantly decreases clearance time, with diminishing returns at higher responder counts.

3.3 Stairwell Count Variation (Scenario B)

To extend the model to multi-floor buildings, we adapt the travel-time component $T_{\text{travel}}(X)$ to reflect the number of available stairwells S . With more stairwells, responders can transition between floors using shorter and less congested paths, effectively reducing edge traversal times in the building graph $G = (V, E)$.

We model this effect by scaling the baseline traversal time:

$$\tau(i, j; S) = \frac{\tau(i, j)}{S},$$

so that

$$T_{\text{travel}}(X, S) = \sum_{(i,j) \in P(X)} \tau(i, j; S) (1 + H(i, j, X)) = \frac{1}{S} \sum_{(i,j) \in P(X)} \tau(i, j) (1 + H(i, j, X)).$$

Keeping the responder count fixed at $R = 2$ and retaining the same inspection and hazard assumptions as in the baseline scenario, we evaluate the model for $S = 1, 2$, and 3 stairwells. The resulting curves show that increasing stairwell count consistently reduces clearance time, with the largest improvement observed when moving from a single stairwell to two. Further increases to three stairwells still reduce $T_{\text{clear}}(X)$, but with smaller marginal gains, reflecting diminishing returns in vertical accessibility.

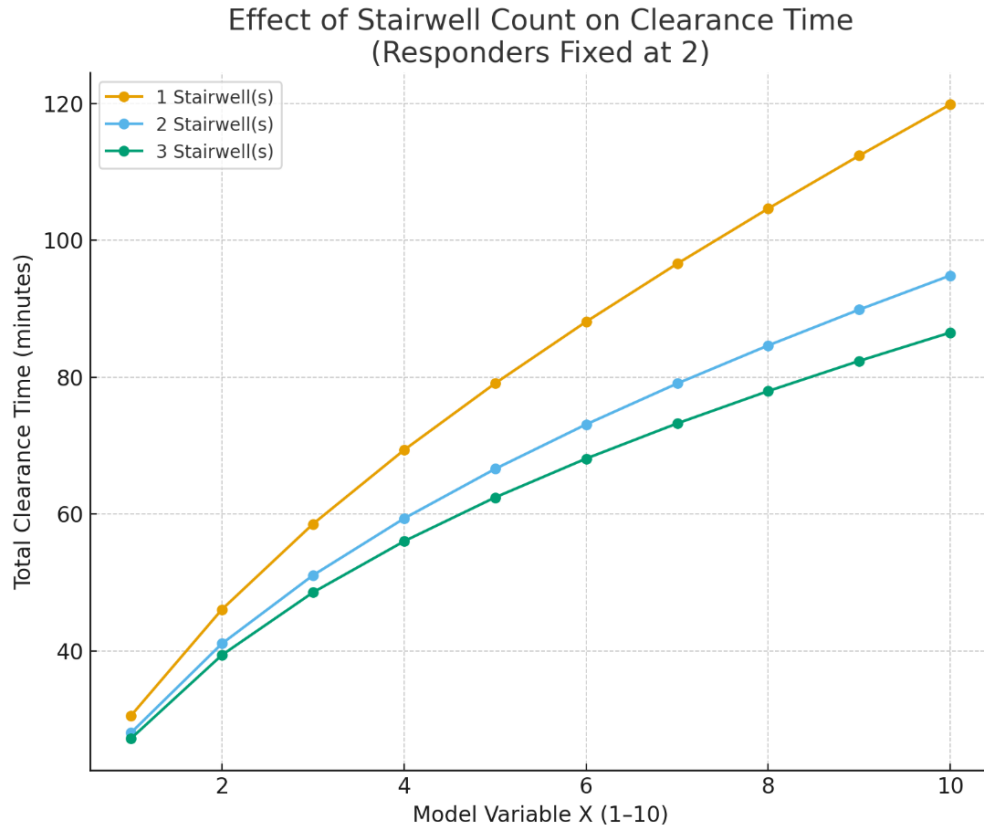


Figure 4: Effect of stairwell count on clearance time with the responder count fixed at $R = 2$. Increasing the number of stairwells shortens vertical travel distances, reducing the travel-time component of $T_{\text{clear}}(X)$ and improving overall sweep performance.

3.4 Summary of Model Behavior for Requirements 2 and 3

Across these applications, the clearance-time model consistently captures the main features of emergency sweeps:

- In the basic one-story office building, the model predicts realistic sweep times for two firefighters under fire-alarm conditions and highlights the influence of central hallway congestion and smoke.

- Varying the responder count shows that $T_{\text{clear}}(X)$ decreases as more responders participate, with diminishing returns at higher responder numbers.
- Adjusting the number of stairwells in multi-floor layouts demonstrates how architectural design directly influences sweep efficiency through its effect on travel time.

These behaviors collectively satisfy the requirements to (i) simulate the baseline sweep scenario, (ii) apply the model to additional layouts with varying numbers of responders, and (iii) showcase how building design parameters and staffing decisions impact the minimum time needed to fully sweep a building.

⁰Figure 2, 3, 4 mislabeled vertical axis in minutes. All results are reported in seconds.

4 Model Extensions Under Realistic Emergency Constraints and Technological Enhancements

4.1 Incorporating More Realistic Emergency Constraints

To extend the clearance-time framework without altering its fundamental structure, we incorporate additional emergency conditions that modify existing variables within the model. These extensions reflect operational realities—such as communication failures, dynamic hazards, and evolving occupant behavior—while preserving the decomposition

$$T_{\text{clear}}(X) = T_{\text{travel}}(X) + T_{\text{inspect}}(X) + T_{\text{redundancy}}(X).$$

Each new emergency constraint modifies exactly one or more of these components but does not introduce new mathematical structures beyond those already established.

Communication Delay or Failure. In emergency conditions involving fire, smoke, or structural instability, communication becomes less reliable. Because room clearance partially depends on receiving confirmation signals, communication delays extend the time during which rooms remain unverified. Responders may rely on physical markings when radio updates fail, effectively increasing redundancy requirements. Thus, communication failures manifest as increases in the redundancy term $T_{\text{redundancy}}(X)$ and indirectly lengthen total clearance time.

Occupants' Awareness of the Emergency. Occupant behavior is already captured through the occupancy probability and its influence on inspection time. In scenarios with low sensory cues—such as odorless gas leaks or distant hazards—evacuation is slower, increasing the likelihood that a room contains an occupant. This raises the expected inspection time for that room,

$$\Delta_r(X) = (1 - p_{\text{occ}}(r, X)) t_{\text{empty}}(r) + p_{\text{occ}}(r, X) t_{\text{occ}}(r),$$

thereby elevating the inspection component $T_{\text{inspect}}(X)$.

Time-Varying Hazards (Fire Spread, Visibility Loss). The hazard slowdown factor integrated into the travel-time term naturally generalizes to dynamic hazards. As fire spreads or visibility decreases, responders encounter progressively worsening traversal conditions. This makes the slowdown factor time-dependent:

$$H(i, j, X) \rightarrow H(i, j, X, t),$$

which increases travel times and may block certain graph edges entirely, forcing longer detours within the building graph $G = (V, E)$. These effects modify only the travel term $T_{\text{travel}}(X)$.

People Moving Dynamically During Evacuation. Evacuee movement introduces congestion that slows responder traversal along shared hallways or stairwells. These conditions adjust traversal times upward along affected edges and may cause responders to select alternate route sets $P(X)$. This is absorbed directly into the travel-time model without structural changes.

Prioritization of High-Risk Areas. High-risk locations—such as laboratories, chemical storage rooms, or childcare facilities—are simply assigned higher priority weights. Responders visit these rooms earlier and may apply more stringent redundancy. The mathematical model remains unchanged; only the inspection ordering and redundancy coefficients within $T_{\text{redundancy}}(X)$ are modified.

Fewer Than Optimal Responders. A reduction in available responders increases the workload per individual. As a result, each responder's path becomes longer, increasing $T_{\text{travel}}(X)$, while inspection and redundancy requirements also intensify. Importantly, the model structure remains intact: only the distribution of tasks among responders changes.

Non-Standard Building Layouts. Irregular or multi-level structures—including dead ends, uneven stairwell placement, and highly compartmentalized areas—affect travel times by requiring longer detours and additional room transitions. Reduced visibility or accessibility may increase redundancy requirements. These effects naturally propagate through the travel and inspection terms of the existing model.

Collectively, these extensions increase realism by modifying parameters already present in the model rather than altering its core composition. Thus, the underlying clearance-time expression continues to govern system behavior, maintaining compatibility with the overall modeling framework.

4.2 Impact of Technology and Recommendations

We now evaluate how modern emergency-response technologies can be integrated into the model by adjusting variables associated with occupancy probability, hazard slowdowns, communication confirmation, and redundancy. These technologies enhance one or more components of the clearance-time function without changing its structure.

Occupancy and Presence Sensors. Real-time occupancy detection reduces the probability that a room contains individuals, lowering expected inspection time $\Delta_r(X)$. These sensors are particularly effective in cellular layouts with many small, enclosed rooms such as offices, childcare centers, or academic wings.

Smoke, CO, and Temperature Sensors. Environmental sensor networks provide earlier and more reliable hazard information, reducing unnecessary traversal of high-risk corridors. More accurate hazard mapping improves routing decisions, decreasing the hazard slowdown factor $H(i, j, X)$. These systems are most impactful in densely connected hybrid or grid layouts.

Automated Room Logging (RFID, QR, Digital Tags). Digitized logging accelerates confirmation and reduces communication-related delays. By shortening the confirmation step associated with each room, such systems directly reduce redundancy time $T_{\text{redundancy}}(X)$. They are well suited for linear and radial layouts where responders naturally split or diverge.

Responder Tracking (UWB, BLE, Indoor Positioning). Real-time tracking reduces spatial uncertainty and prevents accidental omission of rooms without adding inspection duration. This improvement strengthens verification and ensures coverage completeness, especially in irregular or multi-floor buildings.

Building Automation (Smart Doors, Ventilation Controls). Building-automation systems influence hazard behavior directly by controlling smoke movement and improving visibility. These effects reduce hazard penalties and thereby shorten traversal times in the travel component. They are especially useful in large open or hybrid layouts with interlinked spaces.

Together, these technological enhancements interact seamlessly with the existing model components, enabling more accurate, efficient, and reliable emergency sweeps without requiring changes to the model's structural formulation.

5 Letter to the Local Emergency Planning Committee

Dear Members of the Local Emergency Planning Committee,

We are writing to present the results of our evacuation-sweep analysis and to offer practical, building-specific recommendations for improving emergency preparedness across the community. Using a clearance-time framework composed of travel, inspection, and redundancy components—and incorporating hazard progression, communication reliability, and occupant behavior—we evaluated how responders can most effectively perform evacuation sweeps during real emergencies such as fires, gas leaks, and alarm-triggered evacuations.

Our findings emphasize that adaptive planning grounded in building geometry and occupancy characteristics is central to reducing clearance times while maintaining a high standard of safety. By assigning responders strategically, prioritizing high-risk spaces, and integrating technology that enhances verification and route selection, communities can significantly improve the effectiveness and reliability of emergency sweeps. The following sections outline our recommended sweep strategy and provide actionable guidelines tailored to the most common building types in schools, municipal facilities, businesses, and public venues.

5.1 Recommended Evacuation Sweep Strategy

5.1.1 Immediate Floor-Plan Assessment

Responders should begin each sweep with a rapid review of the building layout. Most structures fall into one of six categories—Open, Cellular, Hybrid, Linear, Radial, or Grid—and each layout affects movement, visibility, and room-access patterns.

- Open layouts allow rapid visual clearing of large areas.
- Cellular layouts require systematic room-by-room verification.
- Hybrid, Grid, and Radial layouts demand carefully coordinated routing to avoid overlap and guarantee full coverage.

Identifying the building type early enables responders to choose a sweep pattern that minimizes unnecessary travel and reduces the risk of missed rooms.

5.1.2 Zone-Based Responder Assignment

Before initiating the sweep, the building should be divided into logical zones. Each responder is assigned to a specific zone with an ordered sequence of movement. This prevents duplicated effort and ensures that all critical areas—particularly rooms with high occupancy or hazardous materials—are inspected promptly. In multi-floor buildings, assigning at least one responder per floor is recommended whenever staffing allows.

5.1.3 Early Inspection of High-Risk and High-Occupancy Rooms

Certain spaces should always receive priority during an evacuation sweep, including:

- Childcare rooms, laboratories, workshops, and chemical-storage areas.
- Classrooms, auditoriums, meeting rooms, and other high-density areas.

These rooms present both greater hazard potential and a higher likelihood that occupants may not evacuate promptly. Early inspection improves responder awareness and reduces life-safety risk.

5.1.4 Routing That Accounts for Hazard Development

Hazards evolve over time, especially in fire or smoke conditions. Responders should initially inspect areas closest to the suspected hazard origin when safe to do so, as visibility and traversal conditions typically deteriorate. Stairwells and main hallways—frequent chokepoints during evacuations—should be verified early for accessibility. If hazards block planned routes, responders should shift to alternate paths identified during the initial layout review.

5.1.5 Maintaining Redundancy Without Slowing the Sweep

Redundancy ensures thoroughness, especially in complex layouts or during communication disruptions. We recommend three complementary forms:

- Spatial redundancy: a secondary responder revisits cleared hallways.
- Temporal redundancy: brief rechecks of rooms with uncertain occupancy.
- Priority redundancy: secondary verification for high-risk rooms.

These approaches increase confidence in clearance outcomes while preserving sweep efficiency.

5.1.6 Establishing Clear Communication Protocols

Reliable communication is essential for coordinated sweeps. Responders should confirm cleared rooms promptly. When radio signals fail or are delayed, responders should use visible door markings to indicate status. A designated outside communicator should track progress, manage updates, and provide operational support. These measures maintain coordination even during low-visibility or high-stress conditions.

5.2 Actionable Recommendations by Building Type

5.2.1 Open Layout Buildings (gyms, cafeterias, atriums)

Open spaces provide high visibility and minimal room-to-room traversal.

Recommended practices:

- One primary responder can clear the main area rapidly.
- A secondary responder should inspect adjacent offices, storage rooms, and equipment spaces.
- Redundancy needs are minimal.
- Smoke and environmental sensors can rapidly localize hazards.

5.2.2 Cellular Layout Buildings (office wings, classroom corridors)

These buildings contain numerous small rooms requiring full systematic inspection.

Recommended practices:

- Assign multiple responders to separate corridors.
- Use structured door-marking to avoid missed rooms.
- Apply redundancy primarily to rooms with uncertain occupancy.
- Occupancy sensors can eliminate unnecessary inspections.

5.2.3 Hybrid or Grid Layout Buildings (laboratories, administrative complexes)

Hybrid structures include interconnected rooms, multiple connectors, and large interior spaces.

Recommended practices:

- Divide responders into interlocking sweep zones to avoid route overlap.
- Inspect major corridors and stairwells early.
- Adjust routing as hazards evolve.
- Digital logging systems and environmental sensors greatly improve efficiency.

5.2.4 Linear or Radial Layout Buildings (dormitories, long-corridor facilities)

These layouts often consist of long main corridors with many branching rooms.

Recommended practices:

- Assign responders to separate branches to reduce backtracking.
- Conduct hallway-level redundancy sweeps.
- Monitor central junctions for congestion.
- Real-time responder-tracking improves coordination.

5.2.5 Multi-Floor Buildings

Vertical movement increases sweep complexity.

Recommended practices:

- Assign at least one responder per floor when possible.
- Inspect stairwells immediately for evacuees or hazards.
- Sweep upward or downward depending on hazard origin.
- Use communication equipment designed for multi-level structures.

5.3 Conclusion

Across all building types, the core principles of effective evacuation sweeps remain consistent:

- Plan routes according to building geometry.
- Prioritize high-risk and high-occupancy rooms.
- Apply redundancy selectively and efficiently.
- Integrate technology that reduces inspection time and enhances communication.
- Adjust routes dynamically as hazards evolve and occupants move.

These strategies collectively reduce clearance times while ensuring that both occupants and responders remain safe. We appreciate the committee's ongoing commitment to strengthening emergency preparedness within the community and hope these recommendations support your future planning and operational improvements.

6 Strengths and Weaknesses

Strengths

1. **Flexible Graph-Based Framework.** Representing the building as a graph $G = (V, E)$ allows the model to adapt to multiple layouts, from the HiMCM baseline to multi-floor and irregular structures, without changing the mathematical structure.
2. **Clear Component Decomposition.** Splitting clearance time into travel, inspection, and redundancy components makes the model easier to interpret and analyze, helping identify which factors most strongly affect overall sweep performance.
3. **Realistic Integration of Emergency Conditions.** Communication delays, hazard progression, occupancy uncertainty, and evacuee movement are incorporated by adjusting existing model parameters, improving realism while keeping the model consistent.

4. **Strong Alignment With Problem Requirements.** The model satisfies all parts of HiMCM Problem A, including the baseline sweep, responder variation, stairwell variation, and layout adaptability.

Weaknesses

1. **Limited Empirical Calibration.** Traversal speeds, hazard slowdowns, and occupancy probabilities are reasonable assumptions but not calibrated to real-world firefighter or building data.
2. **Simplified Human Behavior Modeling.** Occupant behavior and crowd dynamics are represented indirectly through penalties, rather than using detailed behavioral simulation.
3. **Path Selection Not Fully Optimized.** Sweep paths $P(X)$ are logically constructed but not optimized using formal graph algorithms or multi-agent routing techniques.
4. **Redundancy Treated Additively.** The redundancy term captures essential rechecks but simplifies more complex responder decisions in high-risk or rapidly evolving emergency conditions.

7 Sensitivity Analysis

We examine how the clearance-time function $T_{\text{clear}}(X)$ changes when its travel, inspection, and redundancy components are individually increased. Here, X represents the baseline sweep parameter used in the model, and the figure compares the baseline curve to scenarios with higher hazard severity, occupancy probability, and redundancy requirements.

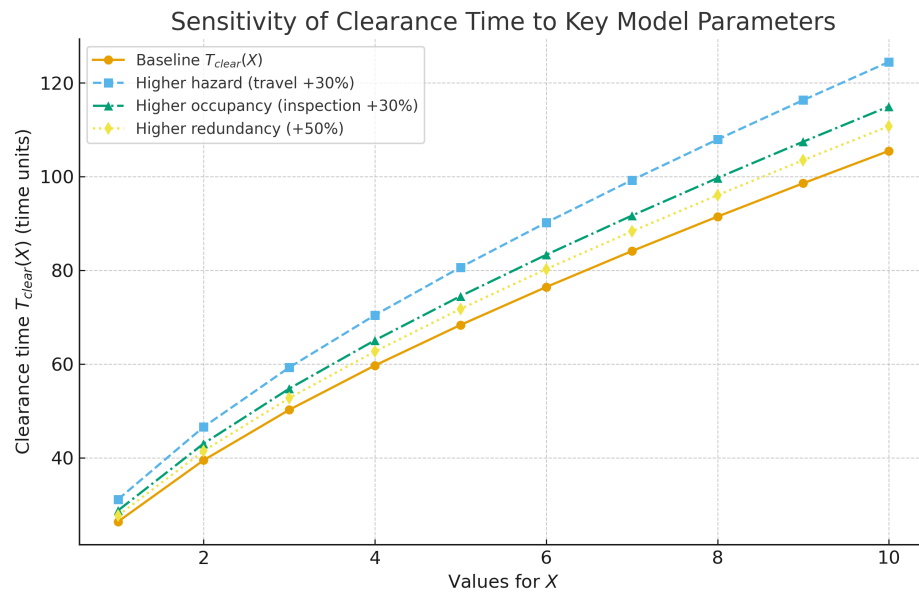


Figure 5: Sensitivity of clearance time to key model parameters. Each curve shows how $T_{\text{clear}}(X)$ changes when one component (travel, inspection, or redundancy) is increased while the others remain at baseline. The x-axis shows “Values for X which represents a general sweep parameter that controls the size or difficulty of the sweep.”

8 Conclusion

Our model provides a unified and adaptable framework for evaluating evacuation sweep strategies across a wide range of building layouts, responder configurations, and emergency conditions. By decomposing clearance time into travel, inspection, and redundancy components, the model captures the fundamental tradeoffs that drive sweep performance while remaining simple enough for practical planning and real-time decision support.

Through baseline analysis and scenario variations, we show that responder availability, stairwell access, hazard severity, and room occupancy all have measurable and predictable effects on total clearance time. Model extensions and technological enhancements further demonstrate how realistic factors—such as communication delays, dynamic hazards, and sensor integration—can be incorporated without altering the model’s core structure.

Overall, the framework satisfies the requirements of HiMCM Problem A by producing a clear, generalizable method for comparing sweep strategies and identifying conditions that most strongly affect evacuation verification. The model supports both operational planning and future improvements in emergency response, offering a structured approach for reducing clearance times while maintaining a high standard of safety.

Report on Use of AI Tools

- [1] OpenAI. (2025). ChatGPT (November 10 version) [Large language model]. <https://chat.openai.com/>
- [2] OpenAI. (2025). ChatGPT (November 13 version) [Large language model]. <https://chat.openai.com/>
- [3] OpenAI. (2025). ChatGPT (November 14 version) [Large language model]. <https://chat.openai.com/>
- [4] OpenAI. (2025). ChatGPT (November 17 version) [Large language model]. <https://chat.openai.com/>

Report on Use of AI Tools

1. OpenAI ChatGPT (Nov 2025 versions, GPT-5)

Query: Can you help me restate Problem A more clearly for the introduction?

Output Used In: Section 1.2 – Problem Restatement.

Query: What is a good name for Section 4 based on the content?

Output Used In: Title of Section 4.

Query: I need help writing assumptions that logically match my model and the HiMCM prompt.

Output Used In: Section 1.1 and Section 1.3 – Baseline Assumptions.

Query: Please convert these assumptions into $\text{\LaTeX}$ using the same format as the paper.

Output Used In: Typesetting in Section 1.3.

Query: Does the paper need a model components summary, and how should it be written?

Output Used In: Introduction of Section 2.

Query: Can you explain whether the occupancy-time equation is consistent with the model?

Output Used In: Section 2.2 – Inspection Time explanation.

Query: Please rewrite this paragraph to make it clearer and more formal.

Output Used In: Sections 1.1, 1.2, and transitions within Section 2.

Query: How do I transition smoothly between the model development section and scenario analysis?

Output Used In: Opening of Section 3.

Query: Can you write a subsection introduction for the building-layout variations?

Output Used In: Section 5 – Letter introduction regarding layout impacts.

Query: Summarize the effect of changing the number of responders in simple text.

Output Used In: Section 3.2 – Responder Count Variation.

Query: Help me write about how communication failures affect my model.

Output Used In: Section 4.1 – Communication Delay or Failure paragraph.

Query: Simplify the hazard slowdown description but keep it technical.

Output Used In: Section 4.1 – Time-Varying Hazards subsection.

Query: Create a strengths and weaknesses section that reflects our modeling choices.

Output Used In: Section 6 – Strengths and Weaknesses.

Query: Make the strengths and weaknesses even simpler.

Output Used In: Final bullet-point version in Section 6.

Query: Write a conclusion that fits the tone of an HiMCM paper.

Output Used In: Initial draft of Section 8 – Conclusion.

Query: Shorten the conclusion further.

Output Used In: Final revised Section 8 – Conclusion.

Query: I need a caption that clearly explains this figure and matches the paper's format.

Output Used In: Figure 1 caption.

Query: I need help producing an AI usage report.

Output Used In: This “Report on Use of AI Tools” section.

Query: Can you create a polished $\text{\LaTeX}$ TikZ diagram showing T_{travel} , T_{inspect} , and $T_{\text{redundancy}}$ flowing into $T_{\text{clear}}(X)$?

Output Used In: Figure 1 – Clearance-Time Decomposition Diagram.

Query: Fix the arrows so they point exactly to $T_{\text{clear}}(X)$.

Output Used In: Final TikZ version of Figure 1.

Query: Can you produce a sensitivity analysis graph using Python?

Output Used In: Figure 5 – Sensitivity Analysis Plot.

Query: Fix the x-axis so it matches my model and make the graph make sense.

Output Used In: Regenerated Figure 5 with corrected x-axis label.

Query: Format the sensitivity analysis figure into $\text{\LaTeX}$.

Output Used In: LaTeX figure environment for Figure 5.

Output Summary: ChatGPT assisted with text editing, formatting, figure preparation, mathematical modeling, scenario analysis, interpretations and clarifying explanations.